Vision-Based Gait Recognition

Detailed process report

Overview

Vision-based gait recognition is a biometric method that identifies a person from the way they walk. Unlike face recognition, which depends on appearance, gait recognition uses motion patterns, posture, and body dynamics. The project is built around a pipeline that starts with raw gait videos, validates and organizes the dataset, selects a suitable camera view, extracts body pose, preprocesses the skeletons, builds a train/validation/test split, trains a model, and finally evaluates the system on unseen test clips.

Dataset organization

The dataset is organized by subject, walking direction, camera label, and trial number. A standard naming convention is used so that each video can be parsed automatically. For example, a clip may be named as subject_direction_camera_trial. This makes it possible to detect valid groups, identify missing files, and automatically match each clip to its subject and view.

Validation and naming rules

Before training begins, the dataset is validated. The system checks whether subjects match the expected list, whether directions are valid, whether the camera labels are recognized, whether trial IDs are in the expected format, and whether the file counts align with the expected structure. Invalid or missing entries are flagged instead of silently included. This prevents the model from learning from broken or mislabeled samples.

Camera selection

Camera selection is one of the most important preprocessing steps. In gait recognition, side-view recordings generally provide the clearest view of the walking pattern because they show lateral movement of the hips and the rhythmic motion of the limbs. The project computes a side-view score for each camera using shoulder ratio and hip travel. The camera with the stronger side-view score is chosen only if it is clearly better. If the two cameras are too similar or the side-view score is weak, the group is left as needs_review instead of being forced into the dataset.

Pose extraction

After selecting valid camera groups, the next stage is pose extraction. A pose-estimation model is applied to each video to detect the subject and localize keypoints such as shoulders, hips, knees, and ankles across frames. These keypoints form a sequence of skeletons that encode the human walking pattern. The model collects 2D coordinates and confidence values for each joint, which are later used as the raw motion representation.

Preprocessing

Raw keypoints are not used directly for training because they may include noisy or missing joints, varying body scale, or inconsistent alignment. Preprocessing therefore standardizes the skeletons by removing invalid frames, aligning the body in a canonical coordinate frame, normalizing scale, smoothing noisy motion, and converting each clip into a temporal sequence. The result is a cleaner representation that preserves the gait pattern while reducing irrelevant variation.

Split design

The split phase decides which valid selected clips go into training, validation, and testing. The split is constructed only from selected and valid groups, and ambiguous groups are excluded. This keeps the evaluation honest. The system also tracks whether a group has a

normal or non-standard number of clips. A clean split helps ensure that the model learns identity from walking dynamics instead of memorizing the same acquisition conditions.

Model training

The processed skeleton sequences are fed into a graph-based gait model. The model uses the relationships between joints and the temporal ordering of frames to learn identity-specific motion patterns. Training is done with mini-batches, and the model is periodically evaluated on validation clips. The best checkpoint is saved based on the validation score. An early stopping mechanism can halt training if validation accuracy no longer improves, keeping the run efficient and reducing unnecessary overtraining.

Evaluation

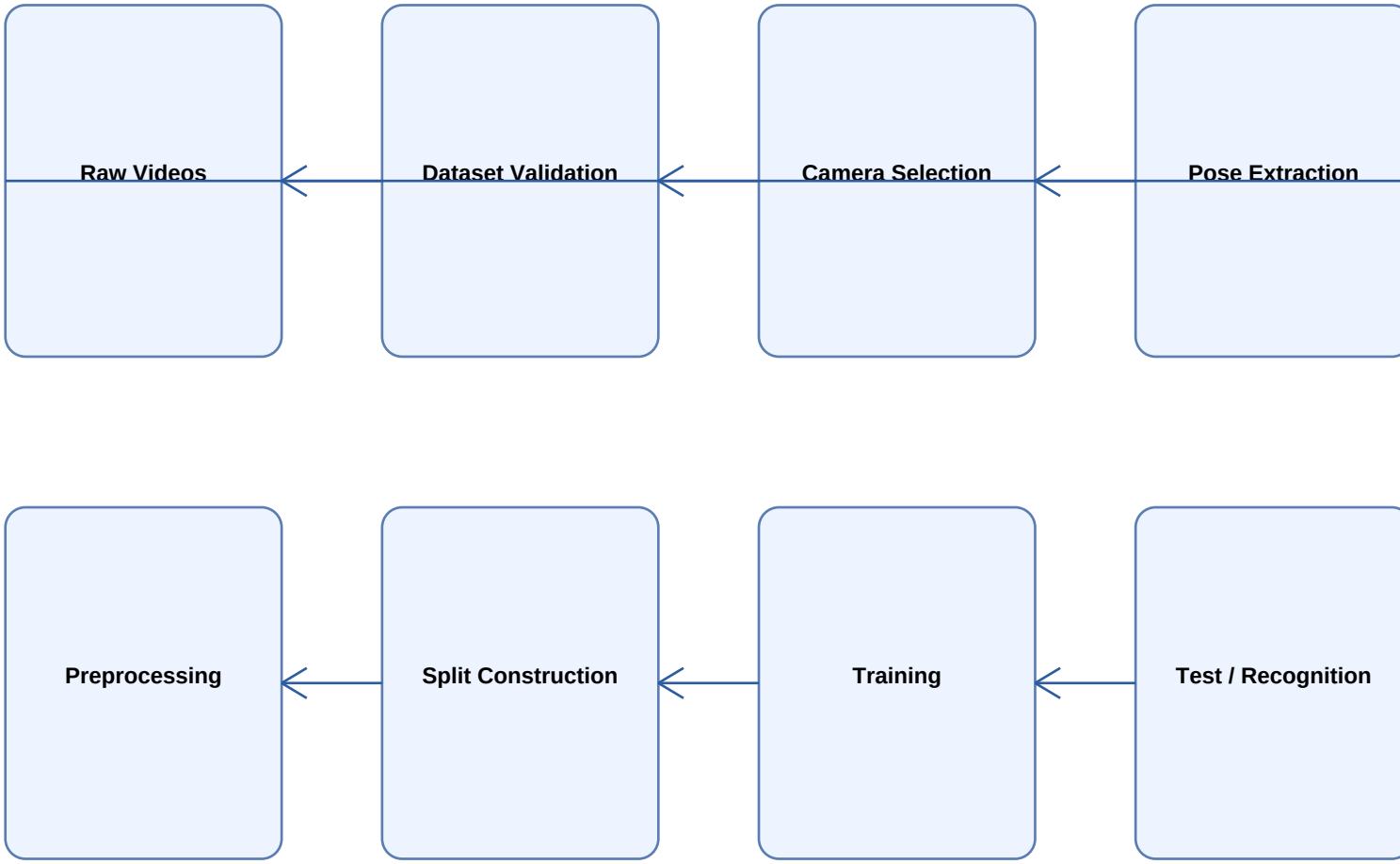
After training, the model is evaluated on the held-out test set. The system compares gallery and probe features, measures how closely each test clip matches the known training gallery, and reports classification metrics such as accuracy, precision, recall, and F1-score. This gives a quantitative measure of how well the gait representation generalizes to unseen samples.

Why the pipeline is conservative

The project deliberately rejects ambiguous camera choices and leaves weak groups out. This is important because a too-aggressive selection policy can produce artificially high accuracy by making the benchmark easier than intended. A conservative pipeline is more reliable because it preserves fairness and reduces data leakage risk. The goal is not to maximize the number of clips at any cost, but to create a benchmark that reflects real gait recognition difficulty.

Conclusion

The vision-based gait recognition workflow is a complete sequence of data validation, side-view camera selection, pose extraction, preprocessing, split construction, model training, and evaluation. Each stage is designed to reduce noise and create a reliable benchmark. The final goal is not only to classify identities, but to do so in a robust and reproducible manner that can be trusted as a realistic gait-recognition pipeline.


Raw Videos
Dataset Validation
Camera Selection
Pose Extraction
Preprocessing
Split Construction
Training
Test / Recognition